

Array

林致宇

Outline

- Arrays
- Character Arrays
- Multidimensional Arrays
- Container

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- Arrays
- Character Arrays
- Multidimensional Arrays

Array

- Store collections of things (e.g., int, double, and so on)
- An array is a container object that holds a **fixed number** of values of a **single type**.
- Why we need arrays?
 - Student List (30 students > 30 variables?)

Declare and Access

資料型別 陣列名稱 [陣列容量];

```
5  int main() {  
6  
7      int iarray[3];  
8      iarray[0] = 1;  
9      iarray[1] = 6;  
10     iarray[2] = 8;  
11  
12     cout << sizeof(iarray) << "\n";  
13     for (int i = 0; i < 3; i++) {  
14         cout << iarray[i] << "\t";  
15     }  
16     cout << "\n";  
17 }
```

Out of Index

```
5  int main() {
6
7      int iarray[3]; // Declare
8      iarray[0] = 1;
9      iarray[1] = 6;
10     iarray[2] = 8;
11     iarray[3] = 10;
12
13     cout << sizeof(iarray) << "\n";
14     for (int i = 0; i <= 3; i++) {
15         cout << iarray[i] << "\t";
16     }
17     cout << "\n";
18 }
```

⚠ Array index 3 is past the end of the array (which contains 3 elements)

Initialization

```
5  int main() {  
6  
7      int iarray1[3] = {1, 6, 8};  
8      int iarray2[] = {2, 5};  
9  
10     for (int i = 0; i < 3; i++) {  
11         cout << iarray1[i] << "\t";  
12     }  
13     cout << "\n";  
14     for (int i = 0; i < 2; i++) {  
15         cout << iarray2[i] << "\t";  
16     }  
17     cout << "\n";  
18 }
```

Initialization (Modern C++)

```
1  #include<iostream>
2  using namespace std;
3
4  int main() {
5      int iarray1[3] {1, 6, 8};
6      int iarray2[] {2, 5};
7
8      for (int i = 0; i < 3; i++) {
9          cout << iarray1[i] << "\t";
10     }
11     cout << "\n";
12     for (int i = 0; i < 2; i++) {
13         cout << iarray2[i] << "\t";
14     }
15     cout << "\n";
16 }
```

std::size() (C++17)

```
1  #include<iostream>
2  #include<array>
3  using namespace std;
4
5  int main() {
6      int iarray1[3] {1, 6, 8};
7      int iarray2[] {2, 5};
8
9      for (int i = 0; i < size(iarray1); i++) {
10         cout << iarray1[i] << "\t";
11     }
12     cout << "\n";
13     for (int i = 0; i < size(iarray2); i++) {
14         cout << iarray2[i] << "\t";
15     }
16     cout << "\n";
17 }
```

Practice

- (Practice 1) Finding the maximum/minimum
- (Practice 2) Find the second minimum
- (Practice 3) Sorting

Practice

- (Practice 4) Given 12345, output 1, 2, 3, 4, 5 (Given one integer number, output 5 integer numbers.)

```
int number = 67395;
```

```
int n5, n4, n3, n2, n1;
```

```
n5 = number/10000;
```

```
n4 = (number%10000)/1000;
```

Outline

- Arrays
- Character Arrays
- Multidimensional Arrays

Character Arrays

```
#include <iostream>
using namespace std;
```

```
int main(int argc, const char * argv[]) {
```

```
    char s1[] = "Hello"; // C-style string
```

```
    char s2[] = {'H', 'e', 'l', 'l', 'o'};
```

```
    cout << s1 << ", " << sizeof(s1) << '\n';
```

```
    cout << s2 << ", " << sizeof(s2) << '\n';
```

```
}
```

```
lincyu@Chih-YudeMacBook-Pro Programs % ./a.out
```

```
Hello, 6
```

```
Hello, 5 // 缺少結束字元 '\0'
```

```
lincyu@Chih-YudeMacBook-Pro Programs %
```

C-style string

- String Length
- String Copy
- String Comparison

String Length

```
#include <iostream>
#include <cstring>
using namespace std;
```

```
int main(int argc, const char * argv[]) {
```

```
    char name[80];
```

```
    cout << "請輸入字串 : ";
    cin.getline(name, 80);
```

```
    cout << "Length: " << strlen(name) << '\n';
```

```
}
```

請輸入字串 : hello

Length: 5

Program ended with exit code: 0

String Copy

```
#include <iostream>
#include <cstring>
using namespace std;
```

```
int main(int argc, const char * argv[]) {
```

```
    char str1[3]= "Hi";
    char str2[6] = "World";
```

```
    cout << str1 << ", " << sizeof(str1) << '\n';
    cout << str2 << ", " << sizeof(str2) << '\n';
```

```
    strcpy(str1, str2);
```

```
    cout << str1 << ", " << sizeof(str1) << '\n';
    cout << str2 << ", " << sizeof(str2) << '\n';
```

```
}
```

```
Hi, 3
World, 6
World, 3
World, 6
Program ended with exit code: 0
```

老師的提醒：
`strncpy(dst, src, sizeof(dst)-1);`

String Comparison

```
5  int main(int argc, const char * argv[]) {
6
7      char str1[] = "Hello";
8      char str2[] = "Hello";
9      char str3[] = "hello";
10
11     cout << boolalpha;
12     cout << (str1 == str2) << "\n";
13     cout << strcmp(str1, str2) << "\n";
14     cout << strcmp(str1, str3) << "\n";
15
16 }
```

實際上是指標!

指標在比較 (恆為 false)

應該用 `strcmp(str1, str2)`

warning: array comparison always evaluates to false

```
[lincyu@Chih-YudeMacBook-Pro Programs % ./a.out
false
0
-32
```

Comprehensive

```
1  #include <iostream>
2  #include <cstring>
3  using namespace std;
4
5  int main() {
6
7      char str1[] = "Hello World";
8
9      cout << strlen(str1) << "\n";
10
11     char str2[80];
12     strcpy(str2, str1);
13     cout << str2 << "\n";
14
15     cout << strcmp(str1, str2) << "\n";
16 }
```

std::string

Convert a C-style String to std::string

```
5  int main(int argc, const char * argv[]) {
6
7      char array[] = "Hello World";
8      string sarray[] = {
9          string(),
10         string(array),
11         string(array, 5),
12         string(array, 6, 5),
13         string(10, 'x')
14     };
15     for (int i = 0; i < 5; i++) {
16         cout << sarray[i] << ", "
17             << sarray[i].size() << ", "
18             << sarray[i].length() << endl;
19     }
20 }
```

Convert std::string to a C-style string

```
1  #include<iostream>
2  #include<string>
3  #include<cstring>
4  using namespace std;
5
6  int main(int argc, const char * argv[]) {
7
8      string str = "Hello World!";
9
10     int len = (int)str.size();
11     char *cstr = new char[len+1];
12
13     strcpy(cstr, str.c_str());
14     cout << cstr << endl;
15 }
```

Concatenation

```
1  #include<iostream>
2  #include<string>
3  using namespace std;
4
5  int main(int argc, char * argv[]) {
6
7      string str1 {"Hello " "World!"};
8      int numberOfBooks = 5;
9
10     cout << str1 << endl;
11     cout << ("There are " + to_string(numberOfBooks) +
12         " books on the desk.") << endl;
13 }
```

Characters

```
5  int main(int argc, char * argv[]) {
6
7      string str1 {"Hello World!"};
8
9      for (int i = 0; i < str1.length(); i++) {
10         cout << str1[i] << endl;
11     }
12     for (int i = 0; i < str1.length(); i++) {
13         cout << str1.at(i) << endl;
14     }
15     str1[0] = 'h';
16     cout << str1 << endl;
17 }
```

Capacity()

可以發現，capacity() 回傳的值會大於字串長度，string 類別設計成一開始就配置一定的儲存空間，已備不時之需。

```
5  int main(int argc, const char * argv[]) {
6
7      string str1 = "Hello World!";
8      string str2 = "";
9
10     cout << str1.capacity() << endl;
11     cout << str1.max_size() << endl;
12     cout << str2.capacity() << endl;
13     cout << boolalpha << str2.empty() << endl;
14
15     str1.reserve(30);
16     cout << str1.capacity() << endl;
17     str1.reserve(5);
18     cout << str1 << " , " << str1.capacity() << endl;
19     str1.resize(5);
20     cout << str1 << " , " << str1.capacity() << endl;
21     str1.resize(30);
22     cout << str1 << " , " << str1.capacity() << endl;
23 }
```

Substring

```
1  #include<iostream>
2  #include<string>
3  using namespace std;
4
5  int main(int argc, char * argv[]) {
6
7      string str1 {"Hello World!"};
8      string str2 = str1.substr(6, 5);
9
10     cout << str2 << endl;
11 }
```

Comparison

- > , >=, < , <=, ==, != (operator overloading)

```
1  #include<iostream>
2  #include<string>
3  using namespace std;
4
5  int main(int argc, char * argv[]) {
6
7      string str1 {"Hello World"};
8      string str2 {"Hello World"};
9      string str3 {"Hello Kitty"};
10
11     if (str1 == str2) {
12         cout << "Equal!" << endl;
13     }
14 }
```

Comparison (compare)

```
5  int main(int argc, char * argv[]) {  
6  
7      string str1 {"Hello World"};  
8      string str2 {"Hello"};  
9  
10     cout << str1.compare(0, 5, str2) << endl;  
11 }
```

String Search

- `find()`
- `rfind()`
- `find_first_of()`
- `find_last_of()`
- `find_first_not_of()`
- `find_last_not_of()`

find()

```
1  #include<iostream>
2  #include<string>
3  using namespace std;
4
5  int main(int argc, char * argv[]) {
6
7      string str1 {"abcdefabcxyzdef"};
8
9      int x = str1.find("abc", 0);
10     int y = str1.find("abc", x+3);
11
12     cout << x << " , " << y << endl;
13 }
```

find()

```
5  int main(int argc, char * argv[]) {
6
7      string str {"Hello World! Hello Kitty! Hello Baby!"};
8      string target = "Hello";
9
10     unsigned int count = 0, pos = 0;
11     string::size_type i;
12
13     while((i=str.find(target,pos)) != string::npos) {
14         count++;
15         cout << "(" << count << " ) " << i << endl;
16         pos = (unsigned int) i + 1;
17     }
18     cout << "NPOS: " << string::npos << endl;
19 }
```

rfind()

```
1  #include<iostream>
2  #include<string>
3  using namespace std;
4
5  int main(int argc, char * argv[]) {
6
7      string str1 {"abcdefabcxyzdef"};
8
9      int x = str1.rfind("def", string::npos);
10     int y = str1.rfind("def", x-1);
11
12     cout << x << " , " << y << endl;
13 }
```

rfind()

```
5  int main(int argc, char * argv[]) {
6
7      string str {"Hello World! Hello Kitty! Hello Baby!"};
8      string target = "Hello";
9
10     unsigned int count = 0, pos = str.length();
11     string::size_type i;
12
13     while((i=str.rfind(target,pos)) != string::npos) {
14         count++;
15         cout << "(" << count << ")" " << i << endl;
16         if (i == 0) {
17             break;
18         }
19         pos = (unsigned int) i - 1;
20     }
21 }
```

find_first_of()

```
5  int main(int argc, char * argv[]) {
6
7      string str {"(1) Hello World! (2) Hello Kitty! (3) Hello Baby!"};
8      string target = "0123456789";
9
10     unsigned int count = 0, pos = 0;
11     string::size_type i;
12
13     while((i=str.find_first_of(target,pos)) != string::npos) {
14         count++;
15         cout << "(" << count << ")" " << i << endl;
16         pos = (unsigned int) i + 1;
17     }
18 }
```

find_last_not_of()

```
5  int main(int argc, char * argv[]) {
6
7      string str = "(02)-1234-5678";
8      string target = "0123456789";
9
10     unsigned int count = 0, pos = 0;
11     string::size_type i;
12
13     while((i=str.find_first_not_of(target,pos)) != string::npos) {
14         count++;
15         cout << "(" << count << ")" " << i << endl;
16         pos = (unsigned int) i + 1;
17     }
18 }
```

insert()

```
1  #include<iostream>
2  #include<string>
3  using namespace std;
4
5  int main(int argc, char * argv[]) {
6
7      string str1 {"Hello!"};
8      string str2 {" World"};
9      str1.insert(5, str2);
10     cout << str1 << endl;
11 }
```

replace()

```
1  #include<iostream>
2  #include<string>
3  using namespace std;
4
5  int main(int argc, char * argv[]) {
6
7      string str1 {"Hello John!"};
8      str1.replace(6, 4, "Bob");
9      cout << str1 << endl;
10 }
```

replace()

```
5  int main(int argc, char * argv[]) {
6
7      string str {"Hi World! Hi Kitty! Hi Baby!"};
8      string ostr {"Hi"};
9      string rstr {"Hello"};
10     string::size_type i;
11     unsigned int pos = 0;
12
13     while((i = str.find(ostr, pos)) != string::npos) {
14         str.replace(i, ostr.length(), rstr);
15         pos = i + 1;
16     }
17     cout << str << endl;
18 }
```

Modify the String Content

```
5  int main(int argc, const char * argv[]) {  
6  
7      string str1 = string();  
8      str1 = "Hello";  
9      str1 += " World";  
10     cout << str1 << endl;  
11     str1.append("!");  
12     cout << str1 << endl;  
13     str1.assign("Hello Kitty");  
14     cout << str1 << endl;  
15 }
```

erase()

```
5  int main(int argc, char * argv[]) {
6
7      string str1 {"Hello World!"};
8      str1.erase(5, 5);
9      cout << str1 << endl;
10
11     string str2 {"Hello World"};
12     cout << str2.erase(5) << endl;
13     cout << str2.erase(1, 3) << endl;
14     str2.clear();
15
16     cout << str2.empty() << endl;
17 }
```

to_string() & stoi

```
1  #include<iostream>
2  #include<string>
3  using namespace std;
4
5  int main(int argc, char * argv[]) {
6
7      int x = 1024;
8      string str1 { to_string(x) };
9      cout << stoi("0xFF", 0, 16) << endl;
10 }
```

stringstream

```
1  #include<iostream>
2  #include<sstream>
3  #include <string>
4  using namespace std;
5
6  int main() {
7
8      int iarr[100];
9      string line, token;
10
11     getline(cin, line);
12     stringstream ss(line);
13
14     int count = 0;
15     while (getline(ss, token, ' ')) {
16         iarr[count++] = stoi(token);
17     }
18     for (int i = count-1; i > 0; i--) {
19         cout << iarr[i] << ' ';
20     }
21     cout << iarr[0] << '\n';
22 }
```

escape R"()"

```
1  #include<iostream>
2  #include<sstream>
3  #include <string>
4  using namespace std;
5
6  int main() {
7
8      string str1 = R("Hello");
9      cout << str1 << endl;
10 }
```

Reference

- <https://cplusplus.com/reference/string/string/>

Outline

- Arrays
- Character Arrays
- Multidimensional Arrays

Multi-Dimensional Array

老師的建議：完全不建議這樣寫

```
5  int main() {
6
7      int array[2][3] = {1, 2, 3, 4, 5};
8
9      for (int i = 0; i < 2; i++) {
10         for (int j = 0; j < 3; j++) {
11             cout << array[i][j] << "\t";
12         }
13         cout << "\n";
14     }
15 }
```

1	2	3
4	5	0

Program ended with exit code: 0

Multi-Dimensional Array

老師的建議：

想成有 2 個長度為 3 的 1 維陣列

```
5  int main() {
6
7      int array[2][3] = {{1, 2, 3}, {4, 5}};
8
9      for (int i = 0; i < 2; i++) {
10         for (int j = 0; j < 3; j++) {
11             cout << array[i][j] << "\t";
12         }
13         cout << "\n";
14     }
15 }
```

Omit the size of the first dimension

- <https://stackoverflow.com/questions/8205148/why-can-we-omit-the-first-dimension-of-a-multidimensional-array-when-we-are-pass>

```
5  int main() {  
6  
7      int array1[][3] = {1, 2, 3, 4, 5};  
8      int array2[2][] = {1, 2, 3, 4, 5};  
9      int array3[][] = {1, 2, 3, 4, 5};  
10  
11 }
```

 Array has incomplete element type 'int []'
 Array has incomplete element type 'int []'

Container

豐富的資料結構集合

Container

- `std::array<T, N>`
- `std::vector<T>`
- Utility
 - `std::pair<T1, T2>`
 - <https://en.cppreference.com/w/cpp/utility/pair>

Container

- `std::array<T, N>`
- `std::vector<T>`

std::array<T, N>

```
1  #include<iostream>
2  #include<array>
3  using namespace std;
4
5  int main(int argc, const char * argv[]) {
6
7      std::array<int, 10> iarray {};
8
9      iarray.fill(8);
10
11     for (int x: iarray) {
12         cout << x << endl;
13     }
14 }
```

std::array<T, N>

```
1  #include<iostream>
2  #include<array>
3  using namespace std;
4
5  int main(int argc, const char * argv[]) {
6
7      array<int, 10> iarray1 {};
8      int iarray2[10];
9
10     iarray1.at(-1);
11     iarray2[-1];
12 }
```

std::array<T, N> (at, front, back)

```
1  #include<iostream>
2  #include<array>
3  using namespace std;
4
5  int main(int argc, const char * argv[]) {
6
7      array<int, 10> iarray1 {};
8      int iarray2[10];
9
10     iarray1.at(-1);
11     iarray2[-1];
12 }
```

std::array<T, N>

```
1  #include<iostream>
2  #include<array>
3  using namespace std;
4
5  int main(int argc, const char * argv[]) {
6
7      array<int, 3> iarray1 {1, 2, 3};
8      array<int, 3> iarray2 {1, 2, 3};
9      array<int, 3> iarray3 {1, 5, 3};
10
11     if (iarray1 == iarray2) {
12         cout << "Equal!" << endl;
13     } else {
14         cout << "Not Equal!" << endl;
15     }
16 }
```

Traditional Arrays

```
1  #include<iostream>
2  #include<array>
3  using namespace std;
4
5  int main(int argc, const char * argv[]) {
6
7      int iarray1[3] {1, 2, 3};
8      int iarray2[3] {1, 2, 3};
9      int iarray3[3] {1, 5, 3};
10
11     if (iarray1 == iarray2) {
12         cout << "Equal!" << endl;
13     } else {
14         cout << "Not Equal!" << endl;
15     }
16 }
```

`std::array<T, N>`

- <https://en.cppreference.com/w/cpp/container/array>

Container

- `std::array<T, N>`
- `std::vector<T>`

std::vector<T>

```
1  #include<iostream>
2  #include<vector>
3  using namespace std;
4
5  int main(int argc, const char * argv[]) {
6
7      vector<int> iarray {1, 6, 8};
8
9      cout << "Size: " << iarray.size() << endl;
10
11     iarray.push_back(5);
12
13     for (int x: iarray) {
14         cout << x << " ";
15     }
16     cout << endl;
17
18     iarray.pop_back();
```

std::vector<T>

- <https://cplusplus.com/reference/vector/vector/>
 - insert

提示：若知道編譯時元素的確切數量，
使用 array，否則使用 vector

Practice

- (Practice 1) Given 27, 35, 16, 8, 92, output 92, 8, 16, 35, 27.
 - Hints: 2 arrays (easier) or 1 array (harder)
- (Practice 2) Try to write a program that call `std::vector<T>.insert()`